

Project Cheap Adapter: one world, different coordinates

A bounded PRH test, converted into deployable linear adapters and a shared multi-encoder Hub.

[Project Resources](#)
[Final Report](#)
[Representation Convergence Defense](#)

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

EXECUTIVE FINDING

The project confirms PRH in a narrow form: trained independent models share mainly local structure that linear maps make useful. It also bounds the claim: convergence is not global or universal, and visible geometry need not transfer.

RESEARCH PROGRAM

8-model core | 14 encoders tested | 2 shipped systems

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.


88.4% of task-trained retrieval

3

Cheap Adapter hub

Seven build it; one joins later

DINOv2 s/b/l, GPT-2, BERT, SBERT and bge-m3 build the 512-d Hub and train the reusable head.


ConvNeXt: 96.7%; Hub/head stay frozen

THE SURPRISE THAT SET UP PART II:

Task-free beats task-trained by 88.4% - evidence about representations, and the motivation for the hub.

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	insufficient: R ² = 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.592

MobileCLIP-to-SigLIP adapter
R²

0.771

bge-SBERT shape rho; all
9,533 items

96.7%

new encoder; no Hub/head
retraining

88.4%

of the jointly-trained
retrieval ceiling

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages, with held-out ConvNeXt at 96.7% of native. Limits: no universal or global identity; scaling is not monotonic beyond the tested range; the source-rank cliff is numerical.

Practical motivation, datasets and model inventory

The engineering constraint and the frozen encoder families behind the experiments, adapters and shared Hub.

FROM SCIENCE TO ENGINEERING

PRACTICAL CONSTRAINT

Limited compute and device memory favor frozen, smaller encoders.

ONE QUERY, BOTH TIERS, ONE RANKING

MobileCLIP -> SigLIP: reuse SigLIP's stronger search space while a constrained device runs only MobileCLIP and a compact linear adapter.

MODEL EXPERT, SHARED SPACE

The Hub connects multiple image and text encoders through one shared coordinate system, allowing a new encoder to reuse existing search heads after fitting only one lightweight entry map.

DATASETS

DATASET	PROJECT USE	EVALUATED	SOURCE
COCO 2017 (train2017)	9,533 image-caption pairs; all encoder caches	9,533 pairs	COCO
Salesforce WikiText-103	GPT-2 / Pythia passages for Experiment A	10,000 texts	HF dataset

VISION MODELS

MODEL	TRAINING SIGNAL	PROJECT ROLE	SOURCE
Apple MobileCLIP-S1	image-text pairs + distillation	phone/source encoder	Hugging Face
Google SigLIP 2 base	image-text sigmoid loss	server/search target	Hugging Face
Meta DINOv2 S/B/L/G	self-supervised visual views	Hub + scale ladder	GitHub
ConvNeXt-base-22k	supervised ImageNet-22k	held-out encoder 8	Hugging Face
FractalDB ResNet-50	FractalDB-1k pre-training	CNN control	GitHub
FractalDB DeiT-tiny	FractalDB-1k pre-training	ViT control	GitHub

LANGUAGE AND TEXT MODELS

MODEL	TRAINING OBJECTIVE	PROJECT ROLE	SOURCE
OpenAI GPT-2 (124M)	next-token prediction	Hub + Experiment A	Hugging Face
EleutherAI Pythia (160M)	next-token prediction	Experiment A	GitHub
BERT-base (110M)	masked language modelling	Hub member	Hugging Face
SBERT all-mpnet-base-v2	sentence-pair contrastive	Hub member	Hugging Face
BAAI bge-m3 (568M)	multilingual retrieval	Hub + text target	Hugging Face

HOW TO READ THIS INVENTORY

The base encoders remain frozen. The learned components are lightweight linear ridge maps and the reusable Hub-to-BGE head. Every blue source label above is clickable.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

PAIRWISE

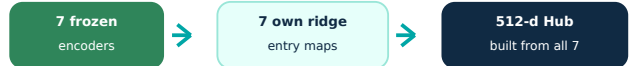
7 encoders -> 42 ordered maps

Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

7 encoders build Hub + get 7 maps

BUILD ONCE: the same 7 encoders define the Hub and get entry maps



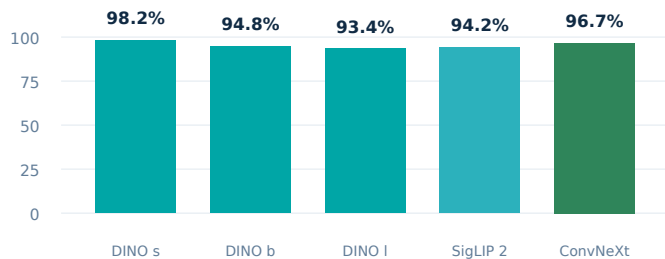
TRAIN HEAD ONCE: DINOv2-small images -> BGE-m3 caption retrieval



LATER: ConvNeXt gets one new entry map; Hub and head stay frozen -> 96.7% of native.

EACH NAMED ENCODER IS THE SOURCE: ENTRY MAP -> HUB -> FROZEN HEAD -> BGE-M3 RETRIEVAL

% of that source encoder's native BGE-m3 R@1 (1,000 held-out items)



WHAT TRANSFERRED

- The same seven encoders both build the 512-d Hub and receive their own entry maps.
- The 512 -> 1024 head trains once on DINOv2-small Hub coordinates, then freezes.
- Add ConvNeXt later by fitting only its entry map - no Hub/head retraining. It reaches 96.7% of native R@1.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.771.

HUB / BESPOKE DIRECT

103.6%

42 ordered pairs (7 x 6). Bespoke baseline: pair-specific ridge A -> B target. Hub: A -> its map -> Hub -> frozen head -> BGE. Mean [Hub R@1 / bespoke R@1] = 103.6%; 5,376 -> 512 whitened PCA can denoise.

LIVE DEMO

39 ms / 0.08 ms

Phone indexing / local search. One query ranks phone and server items together. Adapter: 0.79 MB; 95.3% native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models

only closed-form maps are fitted

Held-out scoring

8,533 train / 1,000 test in the hub

Negative controls

random-map R@1 = 0.001 (chance)

Nonlinear control

MLP ties linear at 0.9183

PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: why shared reality matters

FractalDB tests a key PRH boundary: what structured training contributes without natural photographs, compared with matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Training creates geometry

Fractal training produces small but real shared geometry above matched random twins. Whitening reveals masked CNN structure, and the CNN signal survives PC1 removal.

G10

Geometry is not enough

That visible geometry does not become usable semantic retrieval through the frozen Hub: fractal models remain at the random floor, while ImageNet twins transfer strongly.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN: 0.067 fractal vs 0.036 random. Normalization explains about 10% of the gap; about 90% is training-specific.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

PROJECT VERDICT LEDGER - 34 PRE-REGISTERED PREDICTION UNITS

21

CONFIRMED

Includes learned shared geometry, cross-modal and cross-lineage transfer, and fractal geometry above random.

7

FALSIFIED

Includes the stronger claim that visible fractal geometry must yield semantic transfer; it did not.

4

INCONCLUSIVE

Whether natural-world dependence differs systematically between image and text encoders.

2

NOT MEANINGFUL

Fractal percentage-of-native ratios, because both transfer scores lie below the 0.050 floor.

WHAT THE FRACTAL CONTROL ADDS TO PRH

Fractal training creates shared geometry above random, so convergence is learned rather than purely architectural. But retrieval remains at the random floor while ImageNet twins transfer strongly: shared natural-world experience is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.

More capacity is not monotonically better aligned

Two independent matched ladders rise and then fall, bounding the naive reading that larger models must converge more strongly.

MAIN RESULT

Cross-modal agreement peaks before the largest rung in both vision and text. At the tested scale, more capacity does not guarantee stronger representation convergence.

MATCHED CAPACITY LADDERS - RAW GEOMETRY, NO MAP

DINOv2 IMAGE LADDER

0.381 -> 0.381 -> 0.418 -> 0.339

Small -> base -> large -> giant. Alignment peaks at large, then falls at giant.

CONTRASTIVE TEXT LADDER

0.320 -> 0.352 -> 0.307

bge-m3 -> Qwen-0.6B -> Qwen-4B. The 4B model falls below both smaller rungs.

PRE-REGISTERED VERDICTS

P1 - GIANT HUB TRANSFER INCONCLUSIVE

Measured 100.0% against a 92.9% threshold, but only 2.8 rows per dimension collapsed the denominator. Requires cls pooling or a larger corpus.

P2 - GIANT RHO FALSIFIED

Measured 0.339 with BERT versus a 0.418 threshold. On matched cls+patch pooling, giant remains below large.

P3 / P3B - QWEN FALSIFIED / CONTROL

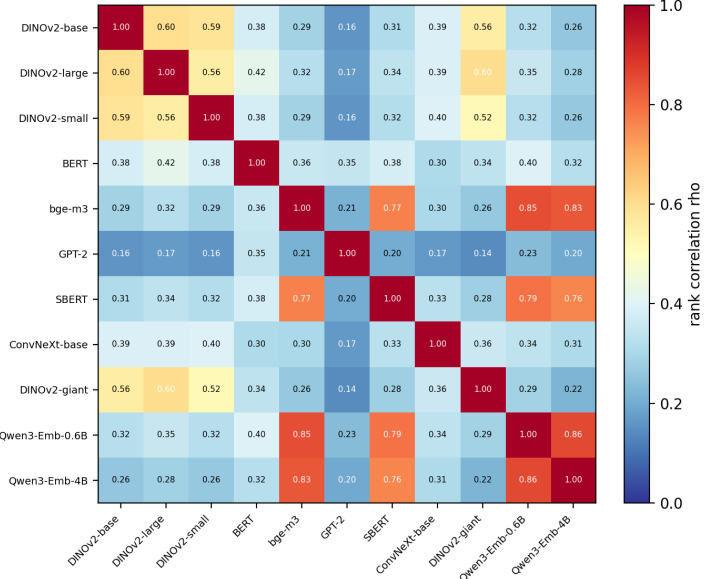
Qwen-4B measured 0.307 versus 0.320. Qwen-0.6B reached 0.352, but at 1.06x bge-m3 it is an architecture/data control, not a scale test.

P4 - RHO VS TRANSFER OPPOSITE DIRECTIONS

Delta rho = -0.079 while transfer rose +7.1 points. Raw shape agreement does not predict recoverable Hub transfer.

55-PAIR SPEARMAN MATRIX - ALL 9,533 ITEMS, NO FITTED MAP

Shape agreement - Spearman of pairwise distances
55 pairs, no map fitted, all 9533 items (+ DINOv2-giant, Qwen3-Emb-0.6B, Qwen3-Emb-4B)



WHAT THE MATRIX ADDS

- Within contrastive text: rho = 0.76-0.86, the strongest pairs in the 55-pair matrix.
- Cross-modal pairs: rho = 0.19-0.35; mean = 0.287, below the earlier 0.290 baseline.
- Pooling correction: giant cls-only rho = 0.238 was invalid for the ladder. Matched cls+patch raises it to 0.339, but the drop remains.
- ConvNeXt 101.2% versus published 96.7% is noise: McNemar p = 0.668; bootstrap 95% CI [96.8%, 106.1%].
- Transfer cell carries a +4.5-point calibration offset. Protocol: rebuilt 512-d H with a frozen 512 -> 1024 head.

Project statistics and PRH confirmation audit

Additional PRH evidence and project-scale statistics from the uploaded audit. Hub, fractal and scale results already shown on pages 1-5 are intentionally omitted.

ADDITIONAL PRH CONFIRMATION EVIDENCE

LEARNED, NOT ARCHITECTURAL
6.3x CKA

Trained models exceed matched random controls; the trained depth curve rises while the random curve decays. Final Report: Exp A.

EXACTLY LINEAR AT TESTED SCALE
0.9183 = 0.9183

MLP adds +0.0000 over ridge. Final Report: F1/F2.

CROSS-MODAL FUNCTION
R@1 0.358

Image-only to text-only retrieval reaches 40.9% of the jointly trained ceiling and 358x chance. Final Report: C.8.

ISOTROPY EXPOSES SIGNAL
102.6%

Whitening rescues collapsed GPT-2 to parity with a contrastive encoder. Final Report: C.11 and C.14.

SELF-IDENTIFYING CORRESPONDENCE
39.7% exact

With zero paired examples, the method identifies exact cross-encoder matches at 595x chance. Final Report: C.12.

NEW BOUNDARIES LOCATED BEYOND PRH

LOCAL, NOT GLOBAL

CKNNA is higher locally for 21/21 pairs, while CCA finds only 6 of 64 strongly shared directions. Hub R-squared is 0.46-0.57 for image spaces but 0.02-0.05 for text spaces.

THE WIDTH CLIFF IS NUMERICAL

The failure step occurs at the source ambient dimension: singular values span 0.005 to 10.79. Truncation or regularisation repairs it, so the cliff is not evidence of a representational boundary.

PROJECT SCALE - NON-DUPLICATED INVENTORY

14

encoders touched

21

named experiments

10

experiment series

~35

Colab notebooks

5

documented corrections

~15

PDF deliverables

DEDUPLICATION RULE
Aggregate verdict counts, the 9,533-item corpus, Hub pair results, held-out ConvNeXt, fractal outcomes and scale-ladder verdicts remain on their original pages and are not repeated here.